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Systems Engineering support for Industry to meet UN SDG

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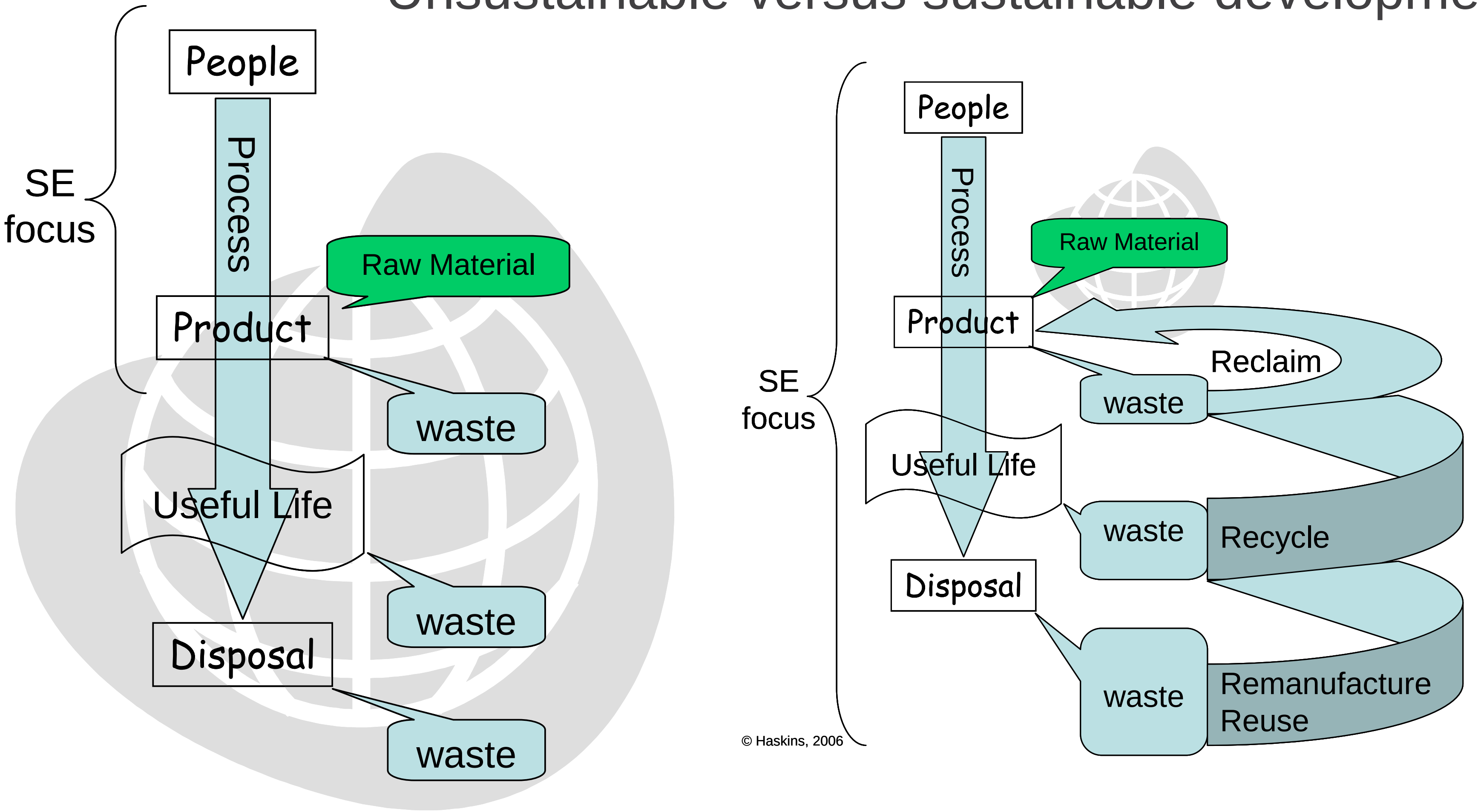
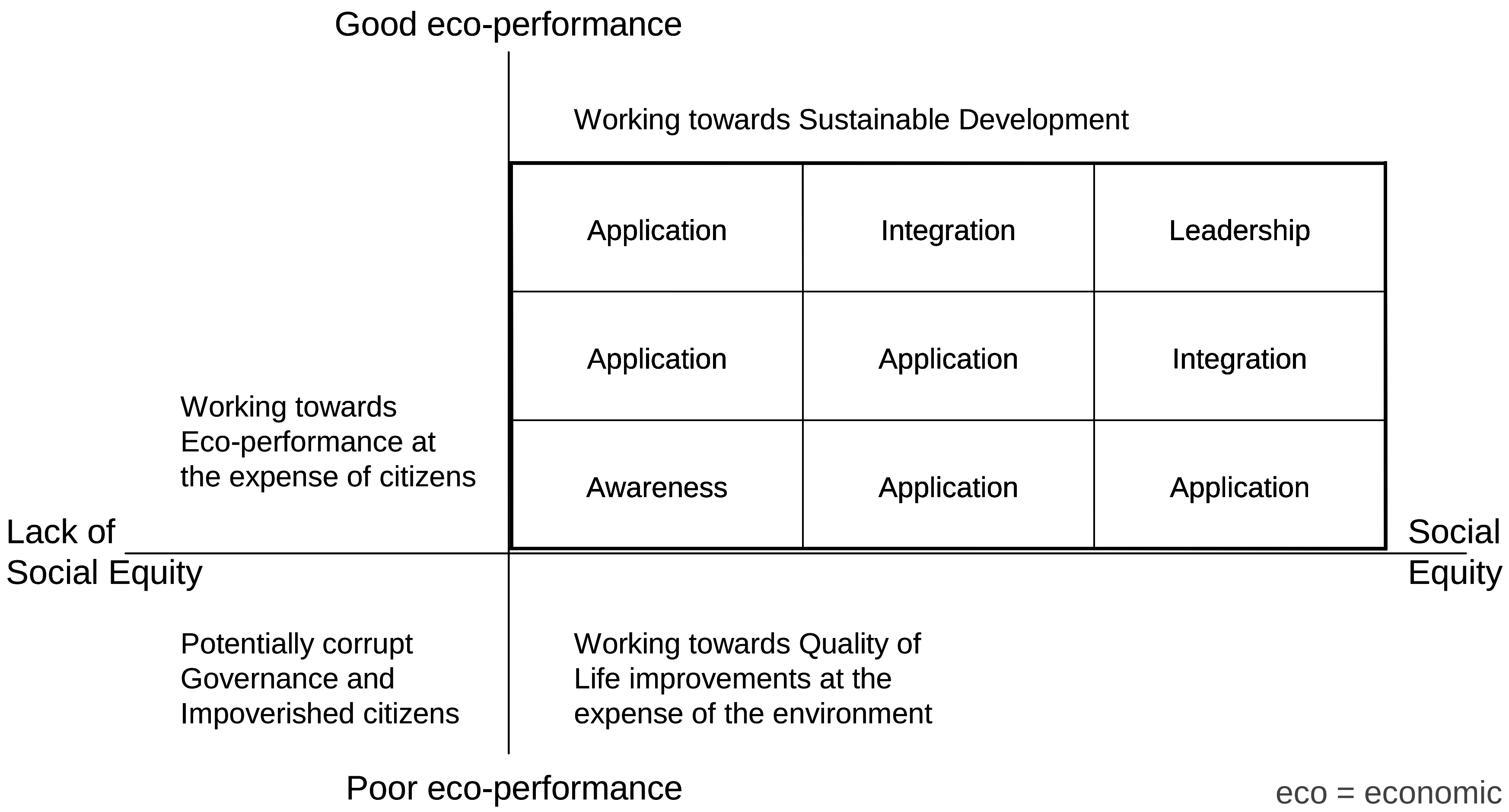
- Expert Systems Engineering Professional
- 20+ years educator
- PhD, NTNU, June 2008: Application of SE to Eco-industrial park development

Technology challenge

Unsustainable versus sustainable development



Matrix of progress toward sustainable development – my PhD case study



eco = economic and environmental

CapSEM project

The initial CapSEM-project, **Capacity building in Sustainability and Environmental Management**, was co-funded by the Erasmus+ Program of the European Union and conducted from October 2016 through October 2019.

Knowledge built during the project has been formally implemented into the curriculum at each partner university. There has been at least one new or adapted bachelor’s or master’s level course developed and tested at each partner from Uganda, Nepal and India.

By structuring sustainability methods and concepts the potentially daunting expanse of topics are taught in a manageable and logical way for practical application and knowledge transfer.

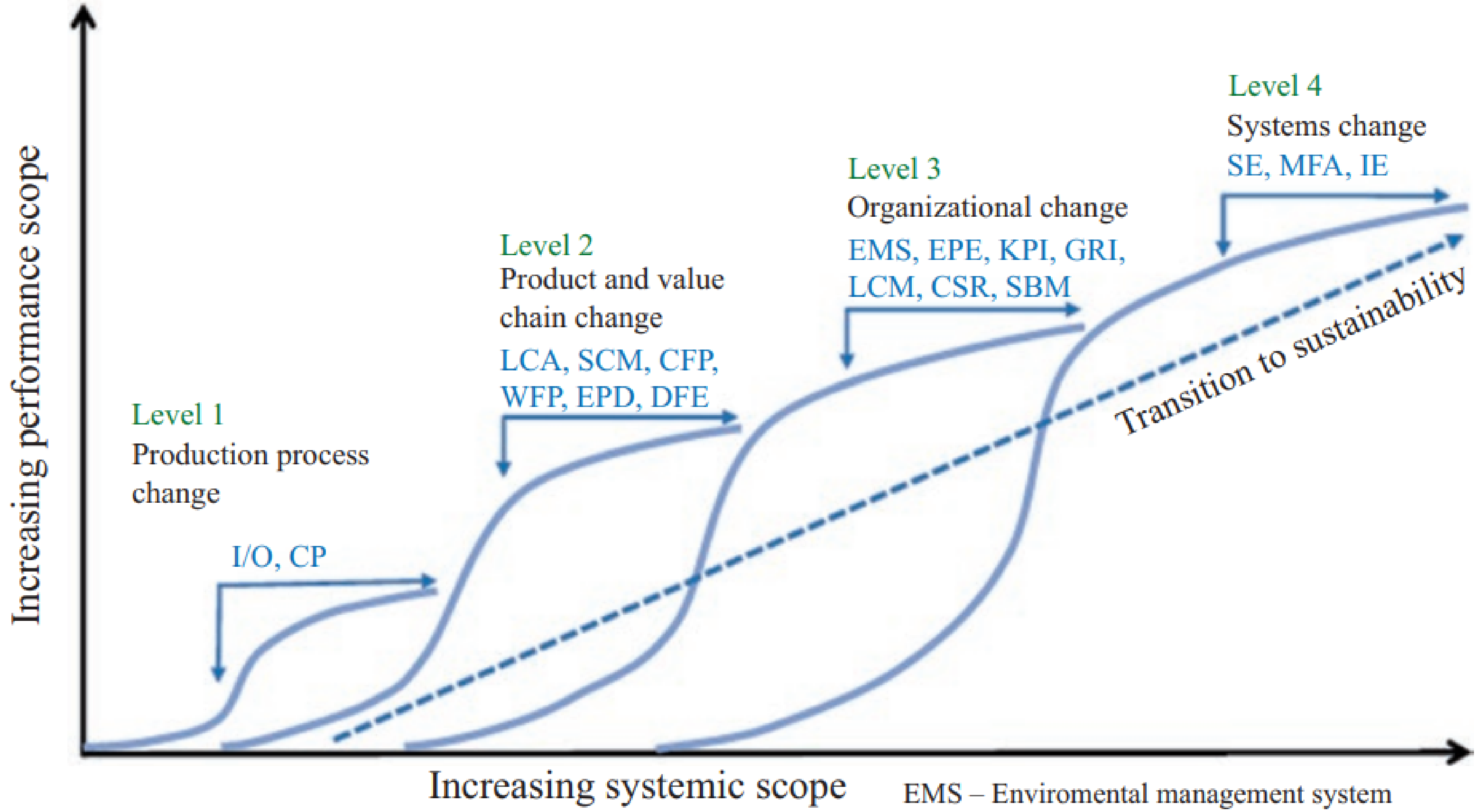
https://capsem.files.wordpress.com/2019/11/capsem-summary-report.pdf

The CapSEM model

- Organizations feel pressure to improve their sustainability performance but may not have either the knowledge or the resources to begin.
- A systematized collection of assessment and management tools for sustainability and environmental management is known as the *Capacity building in Sustainability and Environmental Management* model (the CapSEM Model).
- To streamline their application for the business sector and industry, the methods and tools are positioned in relation to <u>four levels</u> of development:
  - (1) **production** processes,
  - (2) **products** and value chains,
  - (3) **organization** and management and
  - (4) **meta-systems**, for example, industrial sectors or societies

Fet, A.M.(ed) 2023. Business Transitions: A Path to Sustainability: The CapSEM Model (p. 13)
See also https://bh4s.no/toolbox/

The CapSEM model

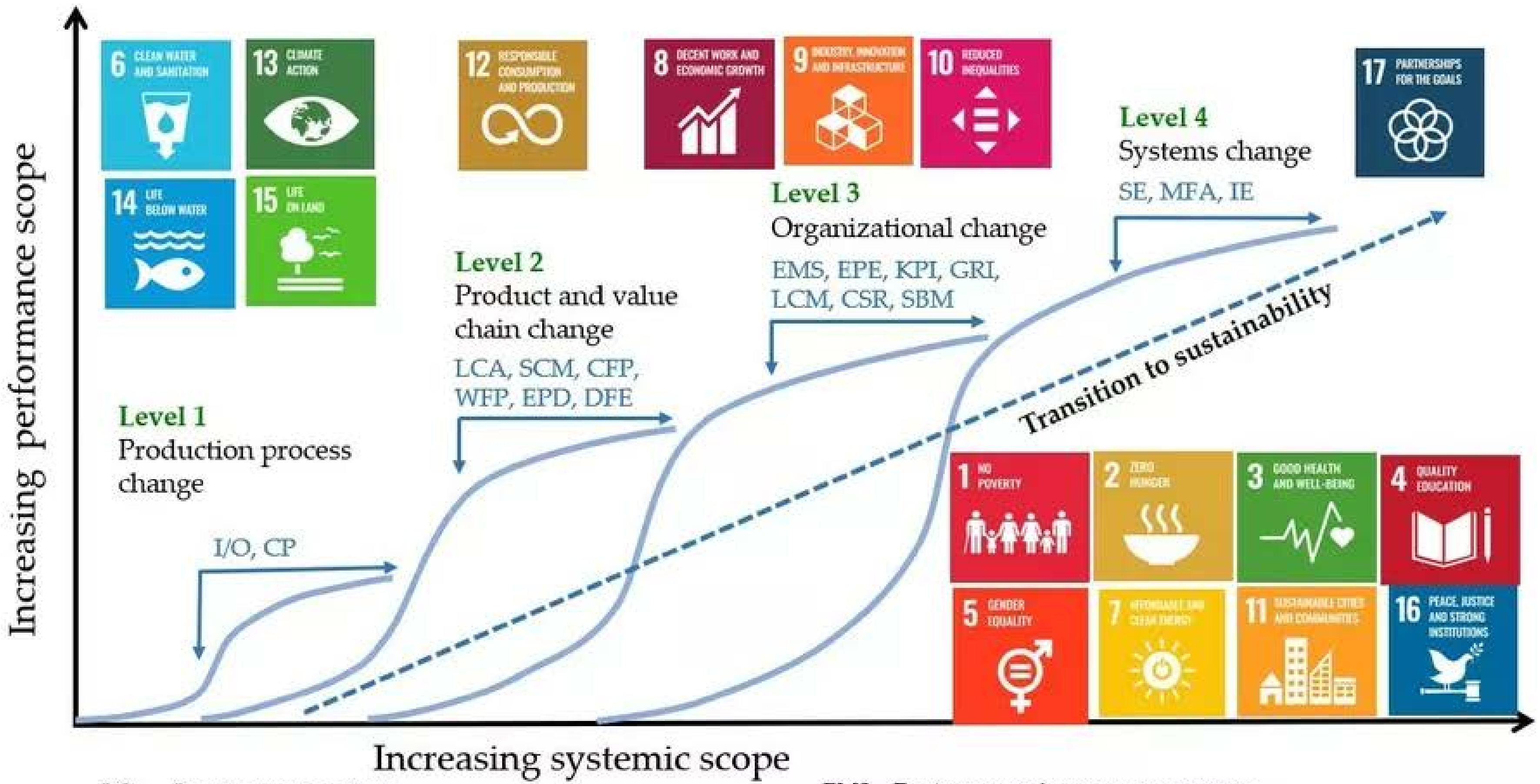


I/O – Input-output analysis
CP – Cleaner production
LCA – Life cycle assessment
SCM – Supply chain management
CFP – Carbon footprint
WFP – Water footprint
EPD – Environmental product delcaration
DFE – Design for environment

EMS – Enviromental management system
EPE – Enviromental performance evaluation
KPI – Key performance indicator
GRI – Global Reporting Initiative
LCM – Life cycle management
CSR – Corporate social responsibility
SBM – Sustainable business model
SE – Systems engineering
MFA – Material flow analysis
IE – Industrial ecology principles

Capacity building in Sustainability and Environmental Management (CapSEM) Model: a systemic approach towards sustainability. (Modified from (Fet and Knudson 2021)

CapSEM mapped to SDG



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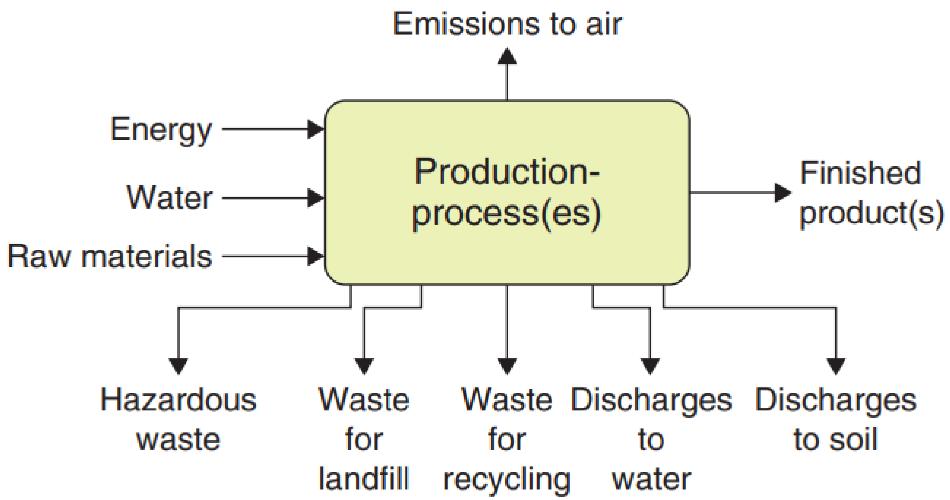
SE enables the CapSEM tools

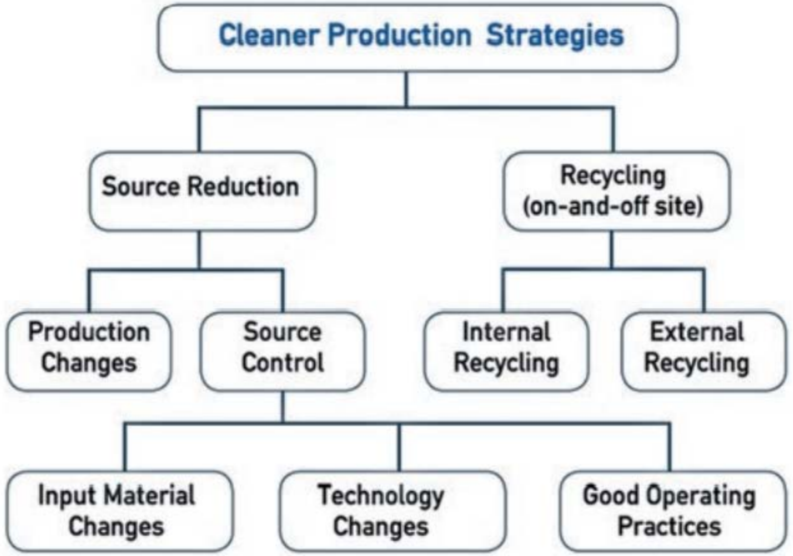
Systems engineering (SE)	Cleaner production (CP)	Life cycle assessment (LCA)	Design for environment (DfE)	Environmental management systems (EMS)	Environmental performance evaluation (EPE)
1. Identify needs	1. Planning and organising	1. Goal and scope definition	1. Needs analysis	1. Environmental policy	1. Commitment
2. Define requirements			2. Requirements	2. Initial planning	2. Planning
3. Specify performance	2. Assessment and preparation	2. Inventory analysis		3. Planning 4. Implementation and operation 5. Checking and corrective action	3. Applying
4. Analyse and optimise	3. Assessment step 4. Feasibility analysis step	3. Impact assessment 4. Interpretation	3. Life cycle strategies and evaluation		
5. Design, solve and improve	5. Reporting 6. Implementation	Application of LCA results	4. Design		4. Reviewing 5. Improving
6. Verify and report			5. Implement	6. Management review 7. Documentation 8. Registration	

Fig. 12.2 Mapping of systems engineering processes to CapSEM Model methods and tools (Fet 2002)

Cleaner Production
«Waste is a failure of sustainability»



Material flow scheme with inputs & outputs for single/series of production process(es)



Principles for CP Strategies from USEPA

Life cycle assessment (LCA)

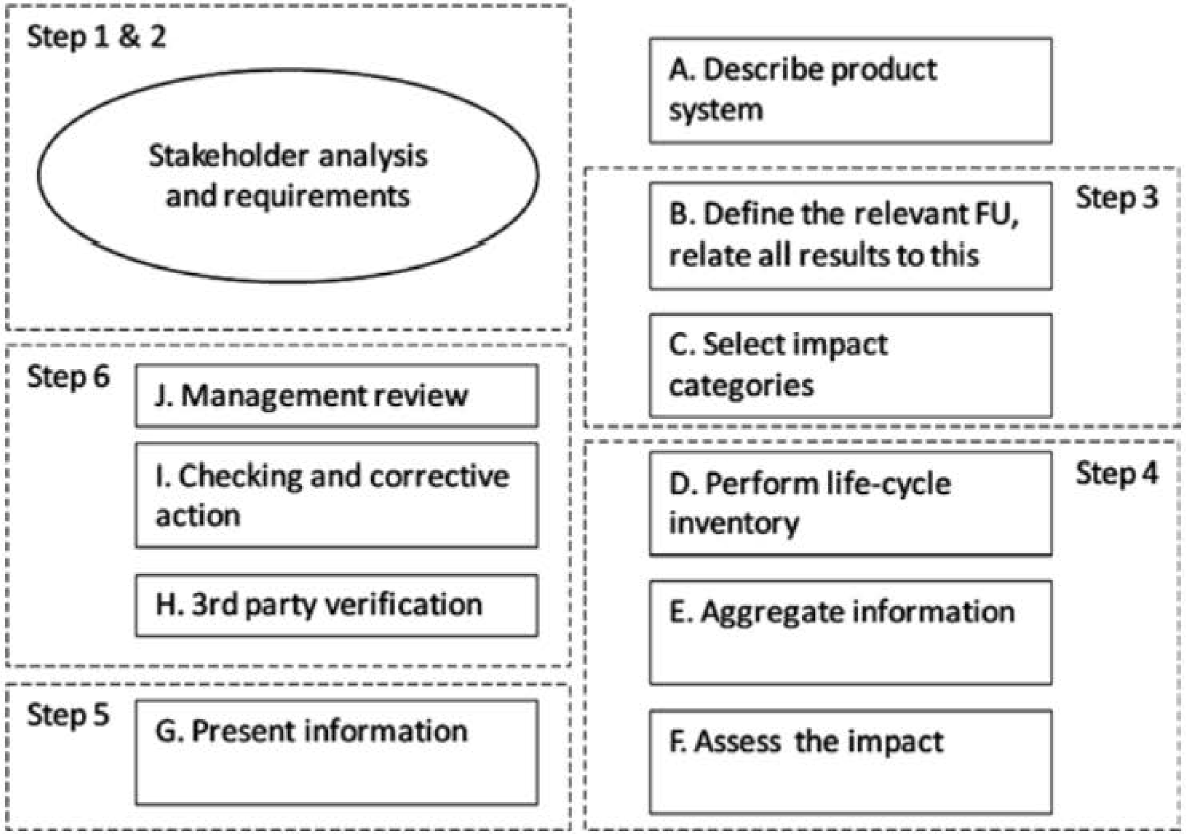


Fig. 14.1 Framework for management and communication of environmental aspects of products. (Skaar 2013)

Design for Environment (DfE)

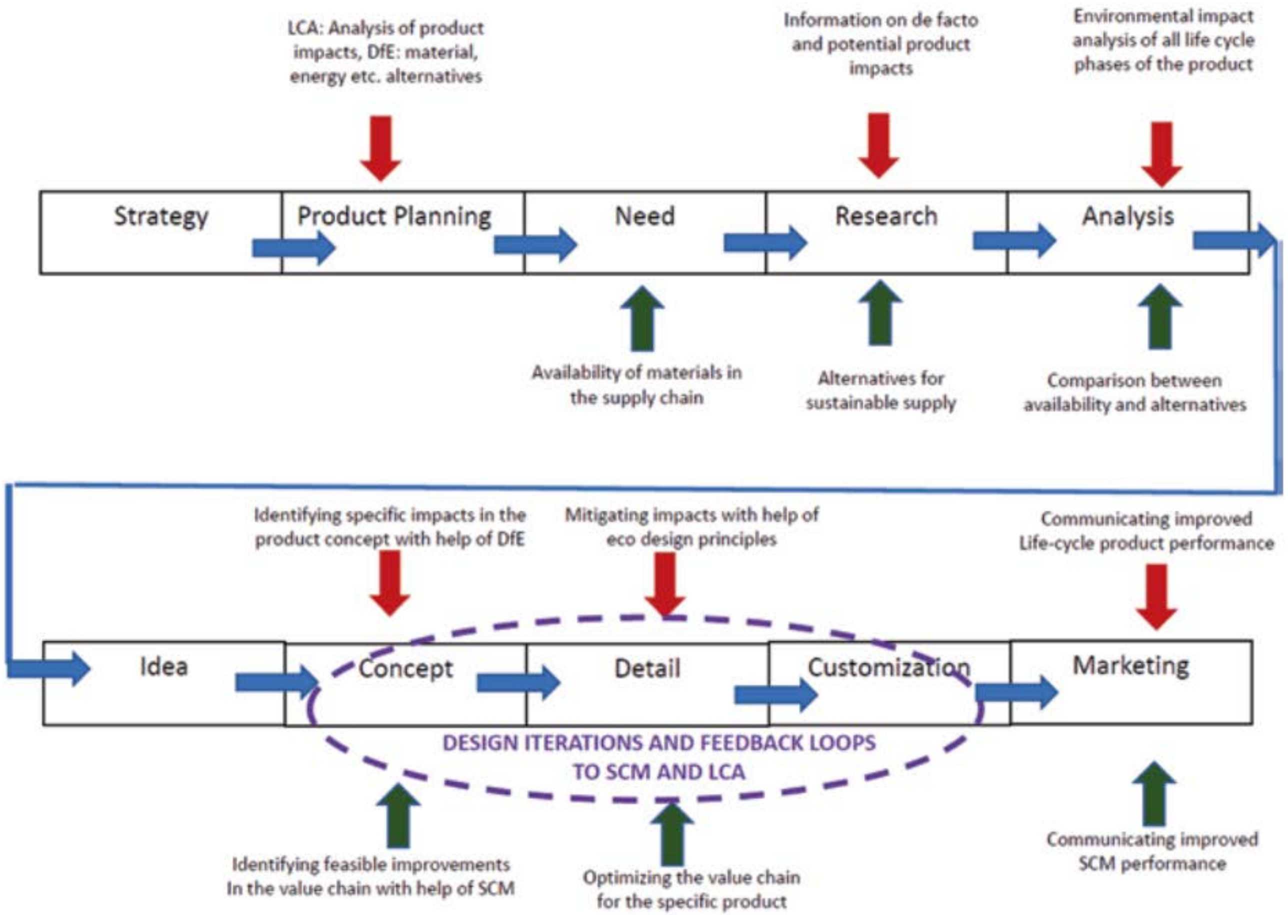


Fig. 5.1 Product development stages integrating aspects of LCA, DfE and SCM

Environmental Management Systems (EMS)

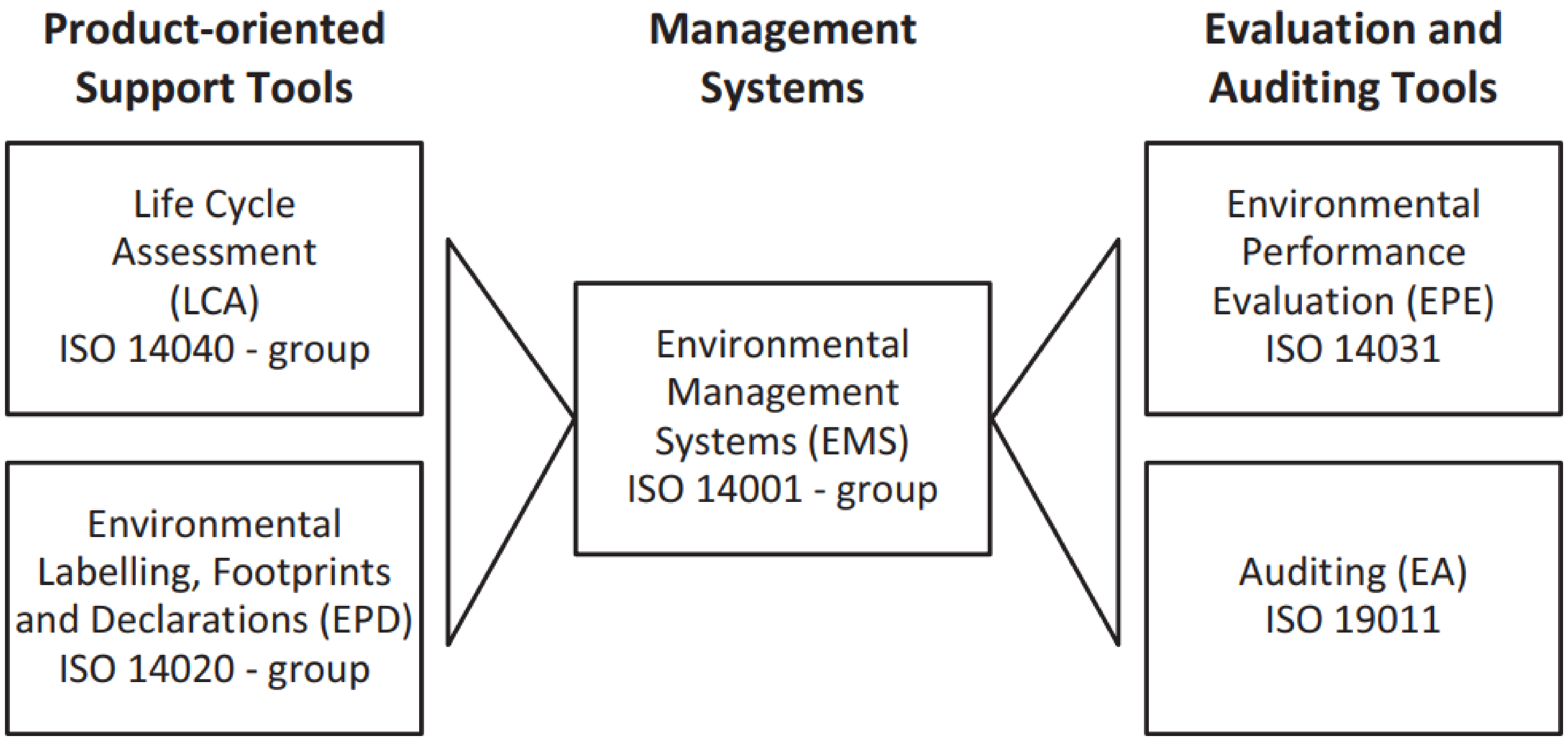


Fig. 7.1 Product-related standards and audit and evaluation standards underpinning environmental management. (Illustrated by examples from the ISO 14000-family)

Environmental Performance Evaluation (EPE) importance of quantification

Table 8.1 Examples of EPIs expressed by OPIs and MPIs

Example of operation performance indicators (OPI).	Example of management performance indicators (MPI).
Category – materials	**Conformance – degree of compliance with regulations**
Quantity of materials used per unit of product	Costs (operational and capital) that are associated with a product's or process' environmental aspects
Quantity of processed, recycled or reused materials used	Return on investment for environmental improvement projects
Category – energy	**Category - financial performance**
Quantity of energy used per year or per unit of product	Costs (operational and capital) that are associated with a product's or process' environmental aspects
Quantity of energy used per service or customer	Return on investment for environmental improvement projects
Category – emissions	**Category - implementation of policies and programmes**
Quantity of specific emissions per year	Number of achieved objectives and targets
Quantity of specific emissions per unit of product	Number of organizational units achieving environmental objectives and targets
Category – wastes	**Category - community relations**
Quantity of waste per year or per unit of product	Number of inquiries or comments about environmentally related matters
Quantity of hazardous, recyclable or reusable waste produced per year	Number of press reports on the organization's environmental performance

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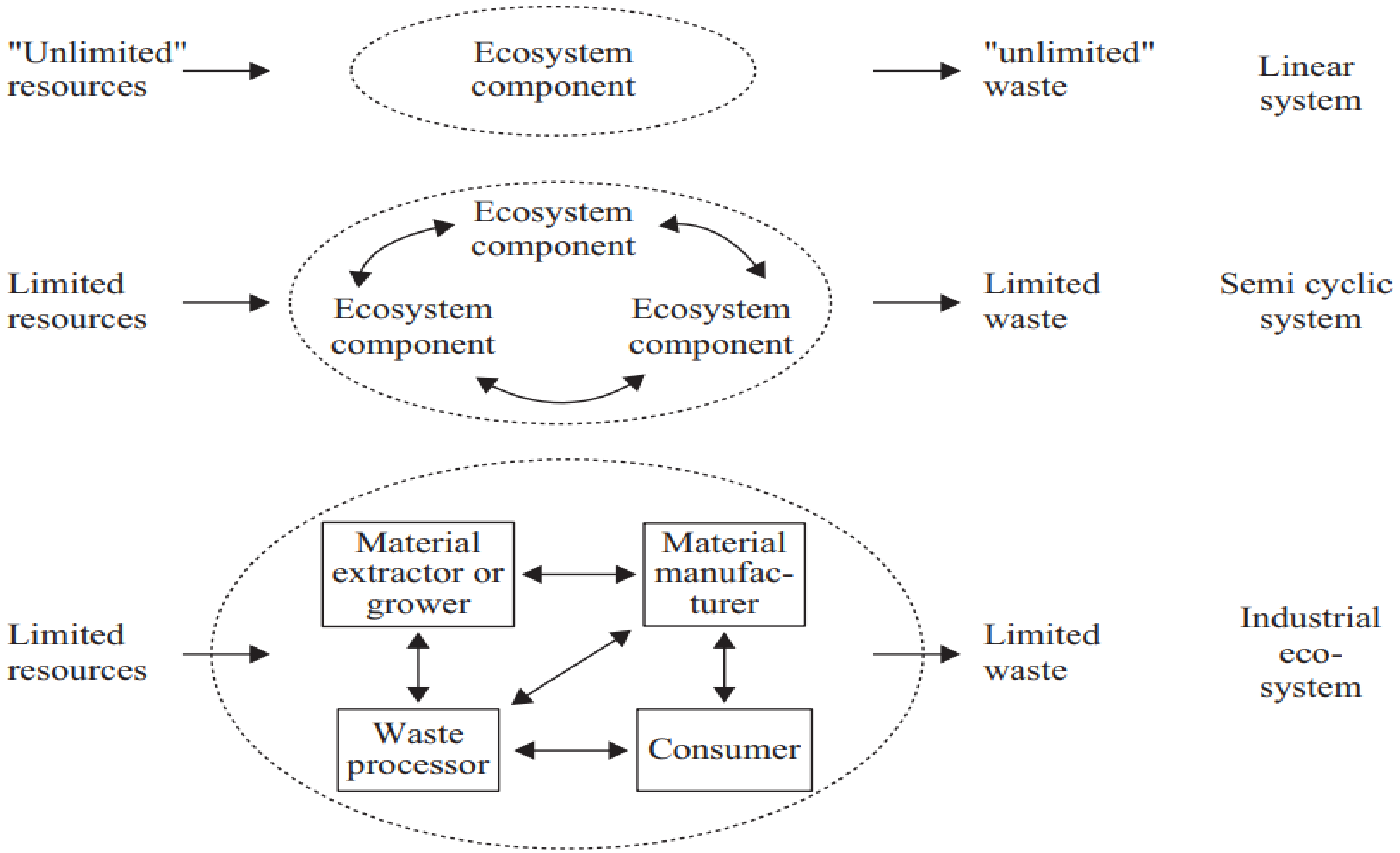
Industrial Ecology (IE)
An ecosystem view of production



Industrial Ecology Characteristics

Table 11.1 Principal characteristics of industrial ecology

	Category
1:	Balancing industrial input and output to natural *ecosystem capacity*, hereunder identifying ways that industry can safely interface with nature in a holistic life cycle perspective.
2:	*Dematerialization* of industrial output, hereunder striving to decrease materials and energy intensity in industrial production.
3:	Creating *loop-closing* industrial practices, hereunder improving the efficiency of industrial processes by re-designing production processes and products for maximum conservation of resources.
4:	Improving *metabolic* pathways for materials use and industrial processes, hereunder create industrial ecosystems by fostering cooperation among various industries whereby the waste of one production process becomes the feedstock for another.
5:	*Systematizing* patterns of energy use, hereunder development of renewable energy supplies for industrial production, and creating energy system that functions as an integral part of industrial ecosystems.
6:	*Aligning policy* to conform with long term industrial system evolution, hereunder adoption of new national and international economic development policies

Modified after Ehrenfeld (1994)

References

- Ehrenfeld JR (1994) Industrial ecology: a strategic framework for product policy and other sustainable practices. In: Green goods: the second international conference and workshop on product-oriented policy, Stockholm, pp 1–40.
- Fet AM (2023) . Business Transitions: A Path to Sustainability: The CapSEM Model (p. 13). Cham: Springer International Publishing. Open Access: https://link.springer.com/book/10.1007/978-3-031-22245-0#about-this-book
- Fet AM (2002) Environmental management tools and their application – a review with references to case studies. In: Conceição P, Gibson DV, Heitor MV, Sirilli G, Veloso F (eds) Knowledge for inclusive development. Quorum Books, Westport, pp 451–464
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- Skaar C (2013) Accountability in the value chain: Product declarations as a tool for measuring, managing and communicating CSR performance. Dissertation, NTNU, Trondheim, Norway

Case study – construction sector

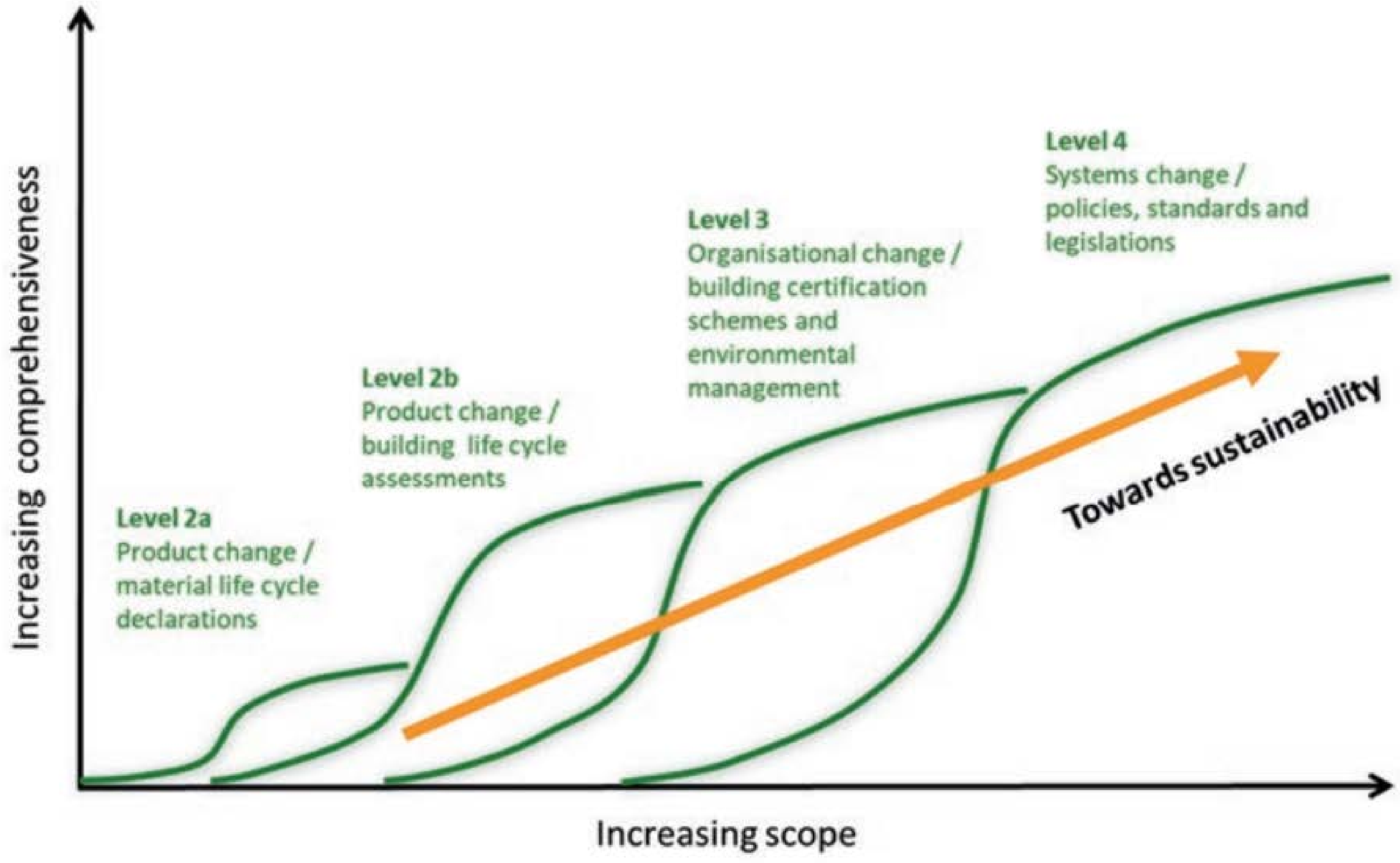
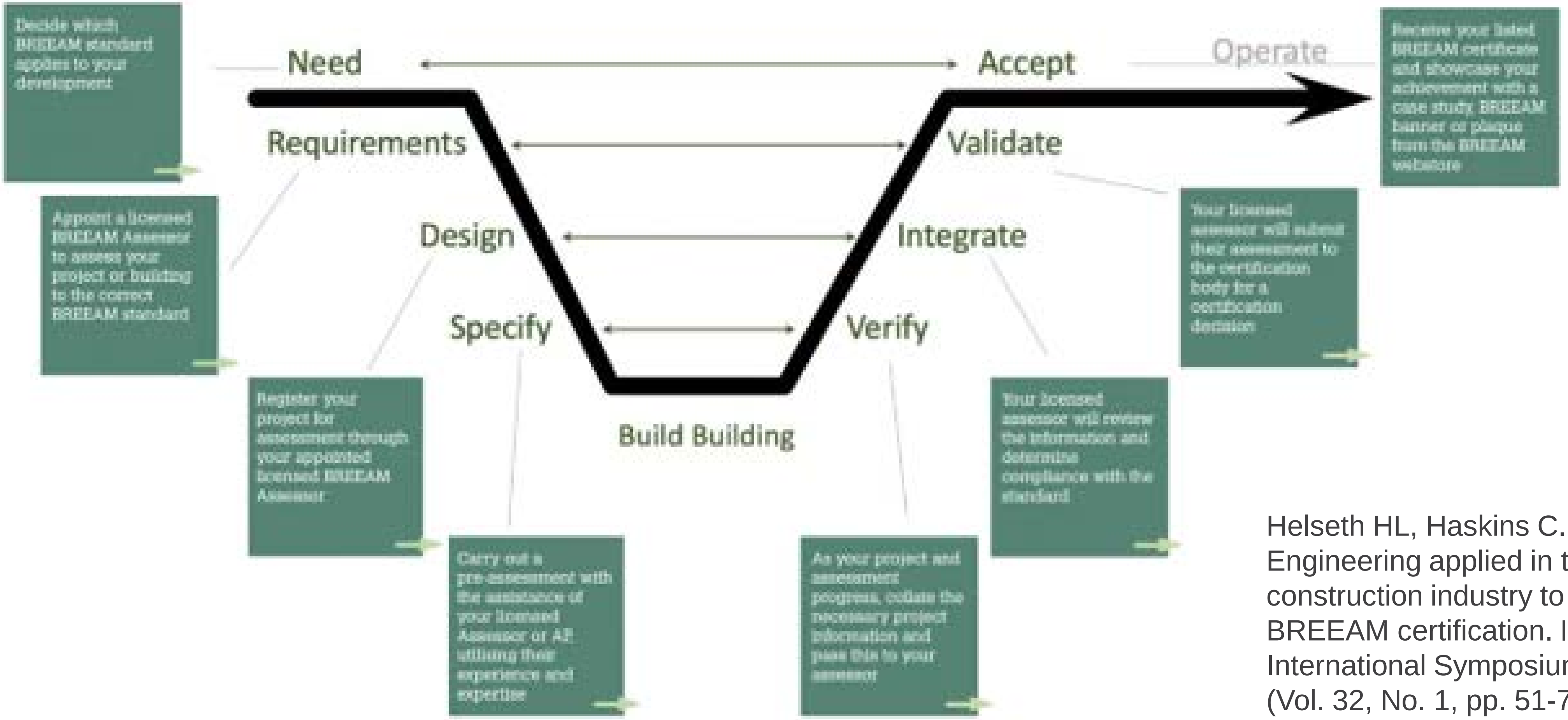


Fig. 16.2 Adapting the CAPSEM Model to the construction sector. (Sparrevik et al. 2021)

Sparrevik M, de Boer L, Michelsen O, Skaar C, Knudson H, Fet AM (2021) Circular economy in the construction sector: advancing environmental performance through systemic and holistic thinking. Environ Syst Decis 41:392–400

BREEAM certification and SE



Helseth HL, Haskins C. Systems Engineering applied in the construction industry to achieve a BREEAM certification. In INCOSE International Symposium 2022 Jul (Vol. 32, No. 1, pp. 51-74).

Case study –
maritime cleaner production, a longitudinal view

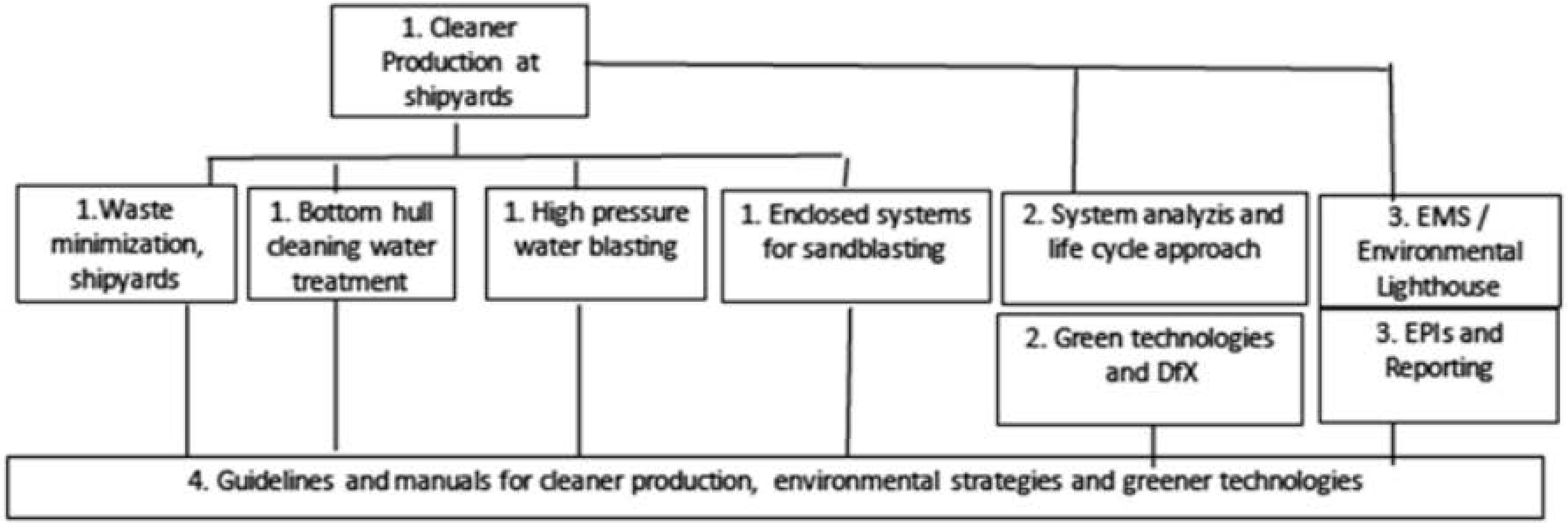


Fig. 18.1 Overview of activities resulting from the Cleaner Production project started in 1993. The numbers in the boxes reflect the Levels in the CapSEM model (modified from Fet 2002a, b)

Case study – using CapSEM to develop a DSS for transportation sector – structuring a MCDM analysis

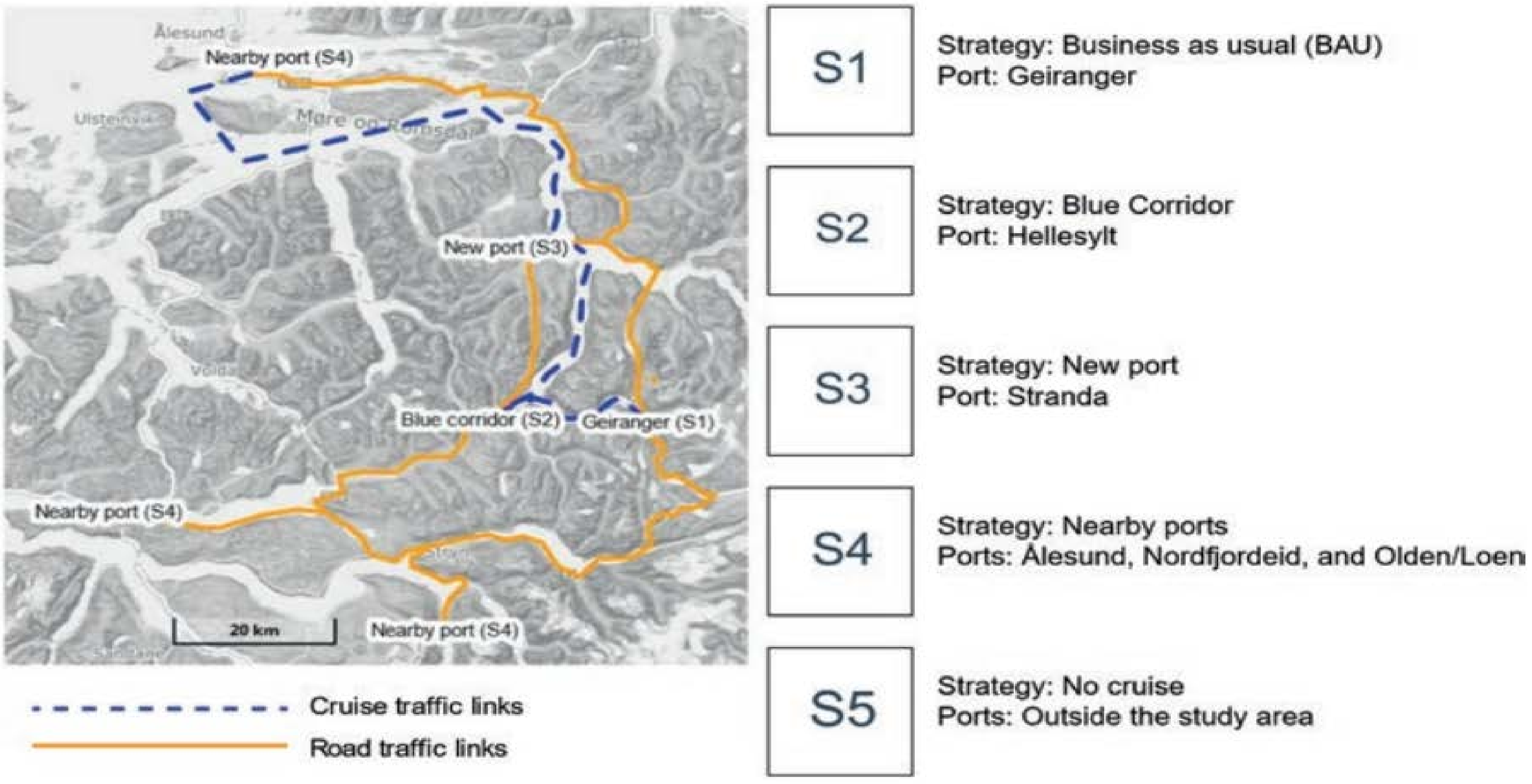


Fig. 19.2 Summary of boundaries and strategies defined in the problem structuring process. The map is created in the Norgeskart portal by ©Kartverket

Why and how systems engineering?

Empirical applications of systems approaches to socio-technical problem domains

- Banathy – designing social systems
- Checkland – soft systems methodology
- Fet – environmental impacts in the marine sector
- Haskins – industrial parks
- Robèrt – strategic sustainable development, backcasting
- Senge – learning organizations, 5th discipline systems thinking
- Warfield – social systems science

Putting SE in the Foreground

- Systems with a human element are **socio-technical** and need **equal focus** on both aspects
- SE is perceived by many as an overloaded framework that only increases overhead and costs
- The real power of SE
  - Systemic – big picture – holistic
  - Systematic – organized way of working
  - Sustainable – solutions must be robust, resilient
- *Practice needs to evolve from linear step-wise processes to apply iterative incremental principles*

A modern SE framework

- Rationale
  - Need to base the modern model on prior knowledge
  - Adjust the message to the non-engineering and management (CEO) audience
  - Achieve an acceptance akin to the PDCA / Deming cycle
  - 4 criteria
    - **Non-linear, express circularity and iteration**
    - **No jargon**
    - **Simple to read and understand**
    - **Applicable to small and medium-sized enterprises and researchers**

Essential SE framework

- SPADE

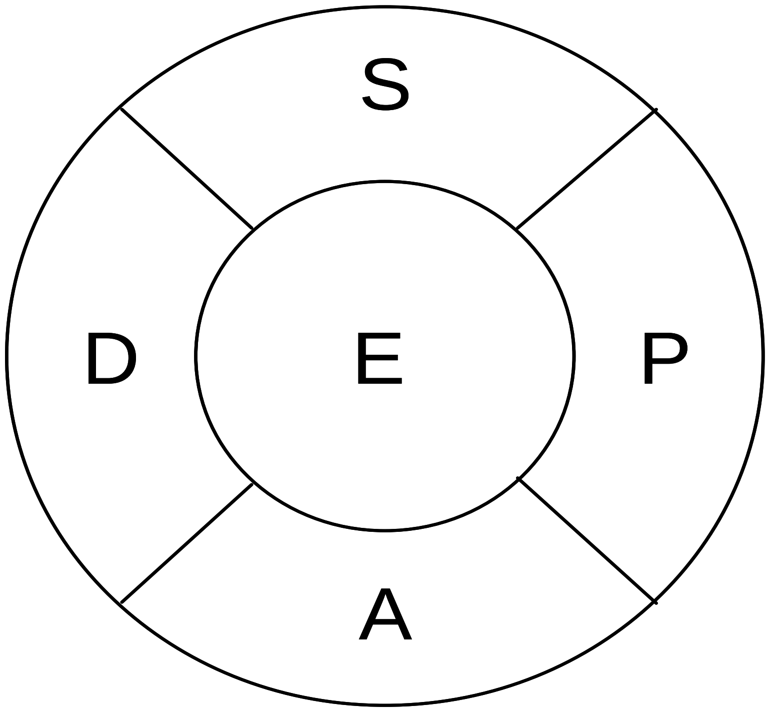
  Stakeholders

  Problem formulation

  Analysis

  Decision-making

  Evaluation



Towards a meta-model of SE practice

- Comprehension - understanding
- Communication – translator services
- Coordination – harmonious functioning
- Cooperation – work toward a goal
- Collaboration – stakeholders, partners
- Continuity – over system life cycle
- Catalyst – champion for change and innovation

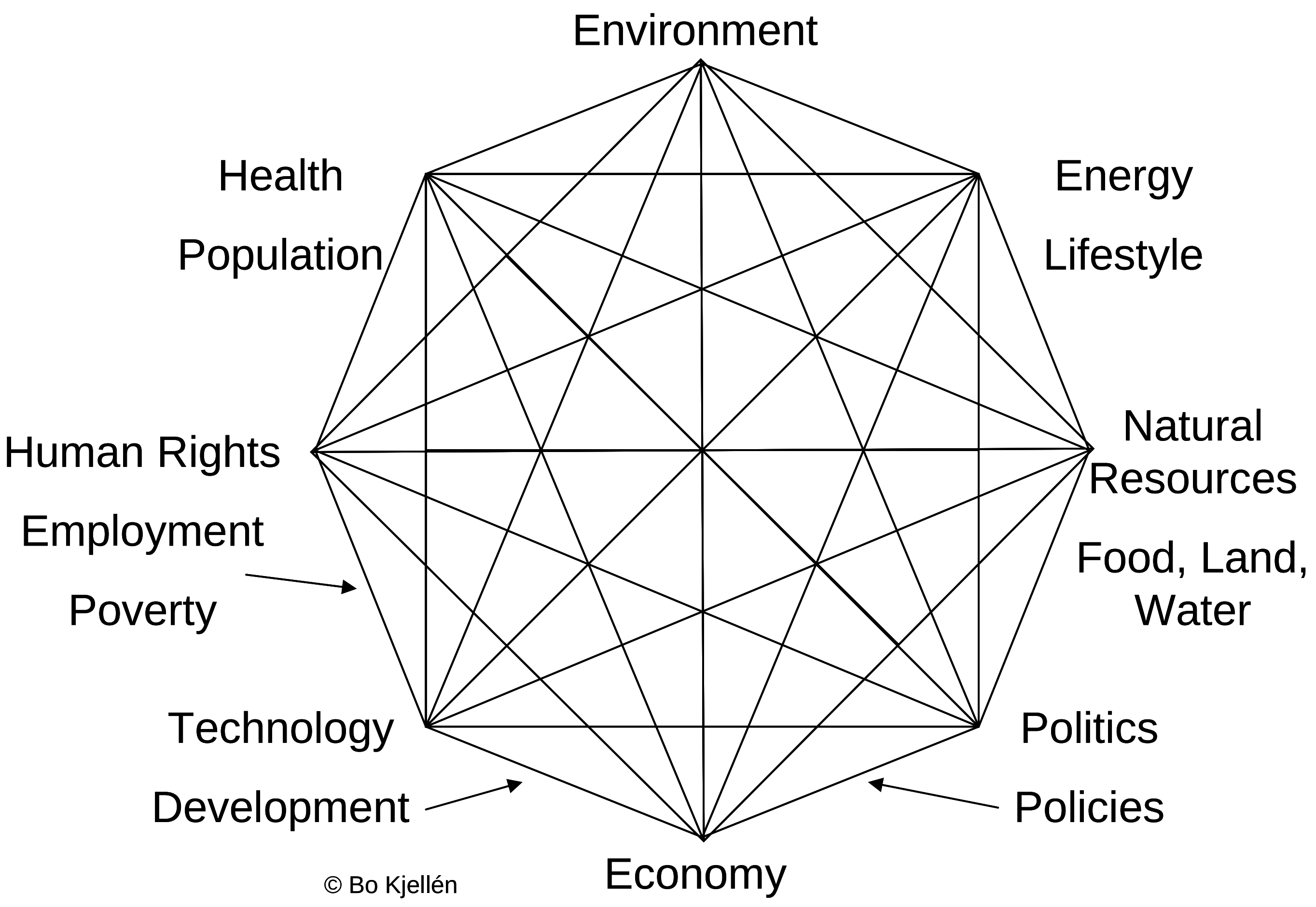
SE is a discipline

- Focus of study - realizing quality systems that meet human needs
- World view or paradigm
- Set of reference disciplines
- Principles and practices
- Active research/theory development
- Deployment of education
- Promotion of professionalism

Educating SE generalists

- Sage recommended
  - team skills, and collaborative, active learning
  - communication skills
  - a systems perspective
  - an understanding and appreciation of diversity; different cultures; business practices
  - integration of knowledge throughout the curriculum a multidisciplinary perspective
  - commitment to quality, timeliness, continuous improvement
  - undergraduate research and engineering work experience
  - understanding social, economic, and environmental impact of **engineering decisions**
  - ethics

Kjellén Diamond of Sustainability



A closing thought

When all the trees have been cut down;
when all the animals have been hunted;
when all the waters are polluted;
when all the air is unsafe to breathe;

only then will man discover he cannot eat money.

Cree Prophecy

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Thank you for your attention!

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